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End Semester Examination JUNE 2025

Name of the Course: ECE

Semester: IVth

Name of the Paper: Microprocessor and Its Application

Paper Code: TEC 403

Time: 3 Hours

Maximum Marks: 100

Note :-

- (i) All questions are compulsory.
- (ii) Answer any two sub questions among a, b & c in each main question
- (iii) Total marks in each main question are twenty.
- (iv) Each question carries 10 marks

Q1	(20marks)	
(a)	Describe the architecture of the 8086 memory segmentation. How does the CPU use segment registers to generate physical addresses?	CO2
(b)	Define a microcomputer system. Explain its main components and their functions.	CO1
(c)	Write 8086 assembly language program to reset parity, zero and interrupt flags.	CO3
Q2	(20marks)	
(a)	Draw and explain the timing diagram for the LDAX B instruction in 8085.	CO2
(b)	Explain any 4 Branching instruction in 8085.	CO3
(c)	Write an 8085 program to find number of 1s in an 8-bit number.	CO3
Q3	(20marks)	
(a)	Explain the functions of various counters in 8254 with specific use. How user can select a counter.	CO5
(b)	Write a program in 8086 to find square root of an 8-bit number.	CO3
(c)	Explain the functions of 8086 maximum mode pins.	CO2
Q4	(20marks)	
(a)	Explain the concept of Effective Address calculation in 8086 with at least three examples using different addressing modes.	CO2
(b)	Write an 8086 Assembly Program to reverse a null-terminated string stored in memory.	CO3
(c)	Explain following instructions in 8086 i) OR [4500], CL ii) MOVSB iii) NEG BX iv) DAS	CO2
Q5	(20marks)	
(a)	Explain interfacing of DAC with 8085 by generating a square waveform.	CO5
(b)	Program 8255 for the following configurations: To get 8-bit data from Port A subtract it from content of port B and display result at Port C.	CO6
(c)	Discuss the priority resolver logic in 8259. How does it handle simultaneous interrupt requests?	CO4

